Technical Test

Q1: Subscriber Usage API

Build a backend service (language/framework of your choice) that lets you record subscriber usage and retrieve what's been recorded. A usage record captures: `subscriberId`, `callMinutes`, `smsCount`, `dataUsageMB`.

Design the endpoints, HTTP methods, and request/response shapes yourself, and document them in your README. Storage can be in-memory. A simple frontend that consumes this API is not required, but will be viewed as a plus.

Q2: Usage Snapshot Automation

Write a cron job that runs 3 times a day (08:00, 12:00, 15:00 WIB), retrieves the current subscriber usage data, and saves it as a CSV file. Choose and document your own file naming convention. Write a separate script that removes CSV snapshots older than 30 days.

Q3: Subscriber & Usage SQL

Given the following subscribers reference table:

ID	Name	Plan	Activation Date
SUB01	Amir	Basic	12-Jan-23
SUB02	Sari	Premium	03-May-22
SUB03	Budi	Basic	20-Sep-24
SUB04	Dewi	Family	15-Feb-21
SUB05	Rian	Premium	08-Aug-23
SUB06	Nia	Basic	30-Nov-24

Assume a `usage` table containing rows like:

subscriberId	callMinutes	smsCount	dataUsageMB	timestamp
SUB01	40	10	1500	2025-08-01 08:00
SUB01	35	8	1200	2025-08-01 12:00
SUB02	90	20	6000	2025-08-01 08:00

SUB02	85	18	5800	2025-08-01 12:00
SUB03	20	5	500	2025-08-01 08:00
SUB04	150	30	9000	2025-08-01 08:00
SUB05	70	15	5000	2025-08-01 08:00
SUB06	25	6	700	2025-08-01 08:00

Write SQL to:

1. Insert a new subscriber named Fajar, Basic plan, activated 24 January 2024.
2. Update Fajar's plan to Premium.
3. Calculate the total data usage (sum of `dataUsageMB` across all snapshots) for all Premium-plan subscribers.
4. Sort and display the top 3 subscribers by total data usage across all snapshots.
5. Write a subquery to find subscribers whose average call minutes per snapshot is less than or equal to 30.

Q4: Troubleshoot & Explain

```
function getTotalUsageMB(records) {
  return records.reduce((total, record) => {
    total += record.dataUsageMB;
  });
}
```

Identify the root cause of the bug, fix it, and explain in writing why it broke and how you would prevent this class of bug in the future.

General Instructions

Submit all code in a Git repository, and include a README explaining how to run the backend application and the automation scripts.

Provide clear documentation for setup and configuration, as a PowerPoint, Word document, or Markdown file (README.md).